

3.5 Carbonyl Compounds

Question Paper

Course	CIEA Level Chemistry
Section	3. Organic Chemistry
Topic	3.5 Carbonyl Compounds
Difficulty	Medium

Time allowed: 80

Score: /60

Percentage: /100

Question 1a

Calcium and its compounds have a large variety of applications.

Calcium metal reacts readily with most acids.

When calcium metal is placed in dilute sulfuric acid, it reacts vigorously at first.

After a short time, a layer of calcium sulfate forms on the calcium metal and the reaction stops. Some of the calcium metal and dilute sulfuric acid remain unreacted.

Suggest an explanation for these observations.

[1 mark]

Question 1b

Calcium ethanedioate is formed when calcium reacts with ethanedioic acid, HOOCCOOH .

Calcium ethanedioate contains one cation and one anion.

i)

State the full electronic configuration of the cation in calcium ethanedioate.

[1]

ii)

Deduce the charge on the cation.

[1]

iii)

Draw the fully displayed formula of ethanedioic acid.

[1]

[3 marks]

Question 1c

Calcium chlorate(I), $\text{Ca}(\text{ClO})_2$, is used as an alternative to sodium chlorate(I), NaClO , in some household products.

i)

The chlorate(I) ion is formed when cold aqueous sodium hydroxide reacts with chlorine.

Write an ionic equation for this reaction. State symbols are **not** required.

[1]

ii)

The chlorate(I) ion is unstable and decomposes when heated as shown.



This reaction can be described as a disproportionation reaction.

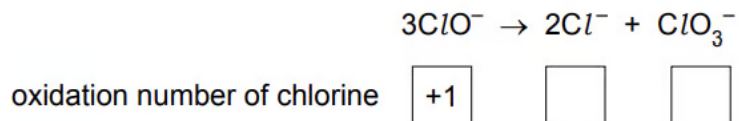
Describe what is meant by disproportionation reaction.

[1]

iii)

Deduce the oxidation number of chlorine in each species for the equation in (c)(ii).

Complete the boxes.



[1]

[3 marks]

Question 1d

Calcium carbonate reacts with 2-hydroxypropanoic acid to form product Y.

2-hydroxypropanoic acid

Y

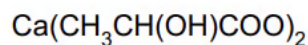
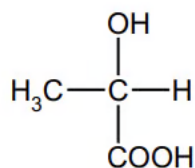


Fig. 1.1

- i)
Identify the **two** other products of the reaction of 2-hydroxypropanoic acid with calcium carbonate.

[1]

Two possible methods of making 2-hydroxypropanoic acid are shown in Fig. 1.2.

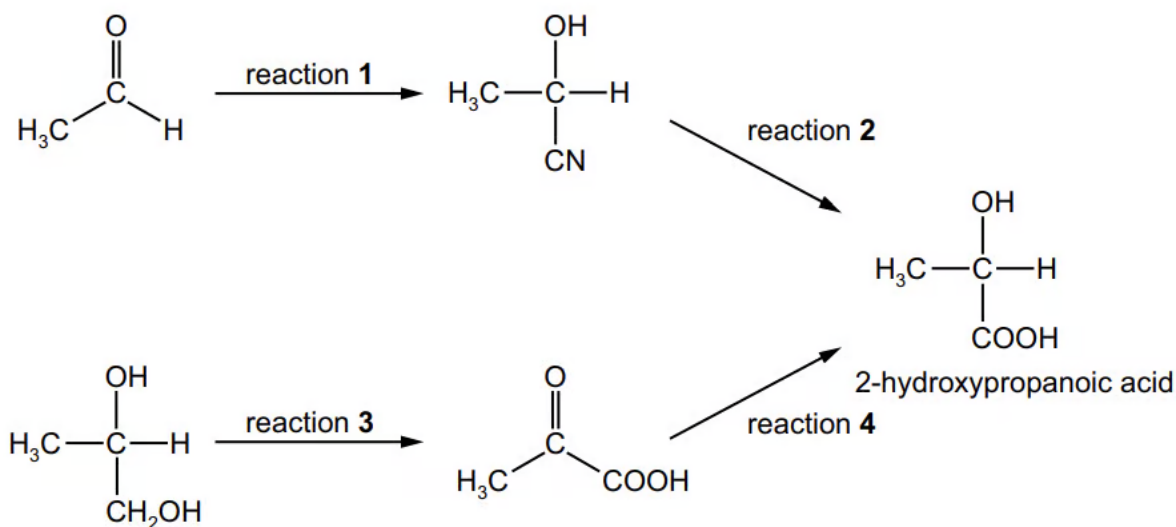


Fig. 1.2

- ii)
State suitable reagents and conditions for reactions **1** and **3**.

reaction **1**

reaction **3**

[4]

iii)

Deduce the type of reaction that occurs in reaction 2.

[1]

iv)

The reagent for reaction 4 is NaBH_4 .Identify the role of NaBH_4 in this reaction.

[1]

v)

2-hydroxypropanoic acid has a chiral centre.

State what is meant by chiral centre.

[1]

[8 marks]

Question 2a

Two isomeric compounds are shown below in Fig. 2.1.

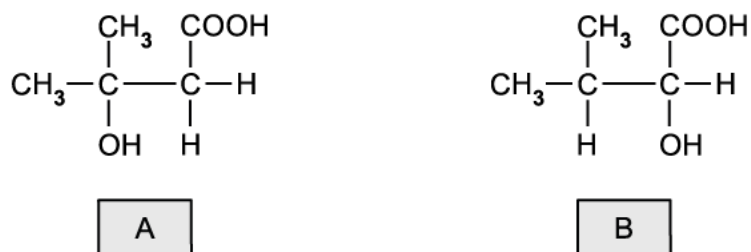


Fig. 2.1

i)

State the name of each isomer.

[1]

ii)

Suggest a chemical reagent to distinguish between these isomers and deduce the type of reaction taking place.

[2]

iii)

State the observations made in each case.

[2]

[5 marks]

Question 2b

i)

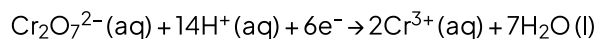
Compound **B**, $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}(\text{OH})\text{COOH}$, can be oxidised into compound **C**.

Deduce the half-equation for the conversion of compound **B** into **C**.

[1]

ii)

The half equation for the oxidation reaction using acidified potassium dichromate(VI) is as follows:



Deduce the overall redox equation for the conversion of **B** into **C**.

[2]

[3 marks]**Question 2c**

The same reaction in part **(b)** can be used to oxidise ethanol into ethanal or ethanoic acid, depending on the reaction conditions.

Outline how the reaction conditions can be changed to produce ethanal or ethanoic acid.

[2 marks]

Question 2d

Under the right conditions, the two molecules of compound **A** can react together to produce a dimer.

i)

Name the type of reaction taking place and state the reaction conditions.

[2]

ii)

Draw the structure of the product, showing clearly how the two molecules are joined.

[1]

[3 marks]

Question 3a

Compound **W** can be converted into three different organic compounds as shown by the reaction scheme in Fig. 3.1.

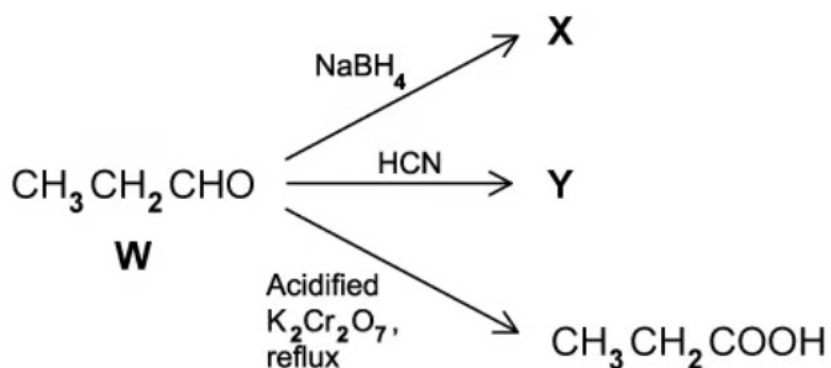


Fig. 3.1

Write an equation for the formation of **X**. Use $[\text{H}]$ to represent the reagent in the equation.

[1 mark]

Question 3b

Identify the specific type of isomerism shown by the product **Y**, in Fig. 3.1.

Draw the mechanism to show the formation **Y** and explain how this leads to the isomers of **Y**.

[5 marks]**Question 3c**

When 5.00 cm^3 of propanal ($M_r = 58.0$) were reacted with an excess of acidified potassium dichromate (VI) solution, 4.25 g of propanoic acid ($M_r = 74.0$) were obtained.

The density of propanal is 0.810 g cm^{-3} .

Calculate the percentage yield for this reaction.

[3 marks]

Question 3d

A different carbonyl compound, **Q**, ($M_r = 72$) reacts with 2,4–dinitrophenylhydrazine but not with Tollens' reagent.

i)

State what you would see when **Q** reacts with the 2,4–dinitrophenylhydrazine reagent

[1]

ii)

State what functional group is present in **Q**.

[1]

iii)

Identify **Q** either by name or by its structural formula.

[1]

[3 marks]

Question 3e

Q can be reduced to compound **R**. For this reaction

i)

State a suitable reducing agent

[1]

ii)

Name the functional group in **R** (two words are required)

[1]

iii)

Give the structural formula of **R**

[1]

[3 marks]

Question 4a

2-hydroxybutanoic acid can be made from propanal by via a reaction with hydrogen cyanide and the subsequent hydrolysis of 2-hydroxybutanenitrile. A possible pathway is shown in Fig. 4.1. The incomplete mechanism for reaction 2 is shown in Fig. 4.2.

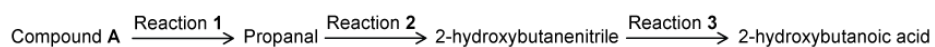


Fig. 4.1

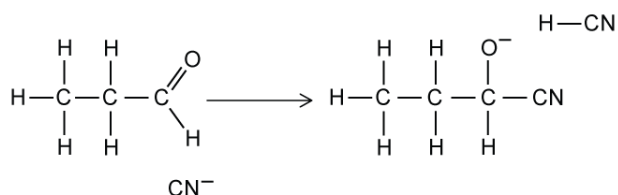


Fig. 4.2

- i)
State the reagents and conditions required for reaction 2.

[3]

- ii)
Complete the mechanism for this reaction.

[4]

[7 marks]

Question 4b

2-hydroxybutanenitrile contains a chiral centre. Draw the three-dimensional structures of the two optical isomers.

--	--

[2 marks]

Question 4c

State the reagent(s) for reaction 3.

[1 mark]

Question 4d

Compound **A** forms propanal when reacted with acidified potassium dichromate. Give the structural formula of compound **A**.

[1 mark]

Question 4e

Compound **A** and propanal were tested using a series of reagents outlined in Table 4.1. For each of the tests listed:

- State the positive result
- Identify which compound(s) would give a positive result

Table 4.1

Reagent	Observation for positive result	Compounds that will give a positive result
$\text{I}_2 / \text{NaOH (aq)}$		
Na_2CO_3		
2,4-DNPH		

[6 marks]

Model Answers: Medium

1a

a) A possible reason why some of the calcium metal and dilute sulfuric acid remain unreacted is:

- CaSO_4 does not react with sulfuric acid

OR

The layer of CaSO_4 prevents the reaction; [1 mark]

[Total: 1 mark]

- The layer of insoluble CaSO_4 prevents the calcium and sulfuric acid from coming into contact with each other
- This means no collisions can occur between the reacting particles and therefore no reaction can occur

1b

b)

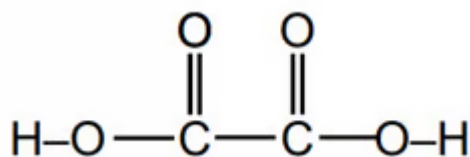
i) The full electronic configuration of the cation in calcium ethanedioate is:

- $1s^2 2s^2 2p^6 3s^2 3p^6$; [1 mark]

ii) The charge on the cation is:

- +2; [1 mark]

iii) The fully displayed formula of ethanedioic acid is:



- Correct formula and all bonds shown; [1 mark]

[Total: 3 marks]

- The cation is the positive ion, which is calcium
- A calcium atom has 20 electrons
- As calcium is in Group 2, it has 2 electrons in its outer shell which it will lose when it reacts
- The resulting calcium ion, therefore, has 18 electrons and a 2+ charge
- When writing the electronic configuration, do not use the abbreviated version $[\text{Ne}]3s^23p^6$ as the question asks for the full electronic configuration
- Count up the number of electrons that you have written in the electronic configuration to make sure they add up to 18
- The calcium ion should have the same electronic configuration as the noble gas, argon
- When drawing the displayed formula, you must show all bonds - students often forget to draw the bond between the O-H and you would not score the mark if you omitted these

1c

c)

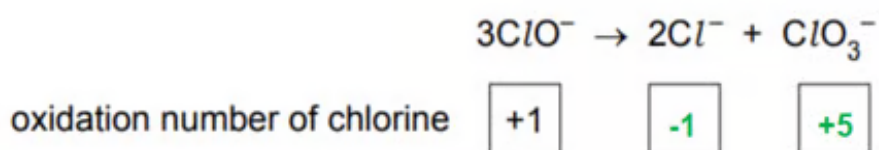
i) The ionic equation for the reaction between cold aqueous sodium hydroxide reacts and chlorine is:

- $\text{Cl}_2 + 2\text{OH}^- \rightarrow \text{Cl}^- + \text{ClO}^- + \text{H}_2\text{O}$; [1 mark]

ii) A disproportionation reaction is one in which:

- A species is (simultaneously) oxidised **AND** reduced
OR
A species (both) gains **AND** loses electrons
OR
• The oxidation number of a species increases **AND** decreases; [1 mark]

iii) The oxidation number of chlorine in each species is:



- -1 **AND** +5; [1 mark]

[Total: 3 marks]

- When chlorine reacts with aqueous sodium hydroxide, the products formed depend on the temperature conditions used
- With cold aqueous sodium hydroxide, sodium chloride, sodium chlorate(I) and water are produced:
 - $2\text{NaOH (aq)} + \text{Cl}_2 \text{ (aq)} \rightarrow \text{NaCl (aq)} + \text{NaClO (aq)} + \text{H}_2\text{O (l)}$
- **NOTE:** With hot aqueous sodium hydroxide, sodium chlorate(V) is produced instead of sodium chlorate(I)
- To write the ionic equation, break down reactants into their respective ions:
 - $2\text{Na}^+ \text{ (aq)} + 2\text{OH}^- \text{ (aq)} + \text{Cl}_2 \text{ (aq)} \rightarrow \text{Na}^+ \text{ (aq)} + \text{Cl}^- \text{ (aq)} + \text{Na}^+ \text{ (aq)} + \text{ClO}_3^- \text{ (aq)} + \text{H}_2\text{O (l)}$
- Cancel out any spectator ions:
 - $2\text{OH}^- \text{ (aq)} + \text{Cl}_2 \text{ (aq)} \rightarrow \text{Cl}^- \text{ (aq)} + \text{ClO}_3^- \text{ (aq)} + \text{H}_2\text{O (l)}$
- The chloride ion, Cl^- is a monoatomic ion, so its oxidation number is the same as its charge, -1
- The chlorate ion, ClO_3^- has the oxidation number of the charge on the ion, -1 and this is the sum of the oxidation numbers of each atom
- Each oxygen atom has the oxidation number -2
- The oxidation number on the ion = oxidation number on chlorine + 3 x oxidation number of oxygen
 - $-1 = \text{O.N. of chlorine} + 3 \times -2$
- So, the oxidation number of chlorine in ClO_3^- must be +5:
 - $-1 = +5 + (3 \times -2)$

1d

d)

i) The **two** other products of the reaction of 2-hydroxypropanoic acid with calcium carbonate are:

- Carbon dioxide
AND
Water; [1 mark]

ii) Suitable reagents and conditions for reactions **1** and **3** are:

Reaction **1**:

- Catalyst / dilute acid; [1 mark]
- KCN / NaCN; [1 mark]

Reaction **3**:

- Reagent: potassium dichromate(VI); [1 mark]
- Reaction conditions: H_2SO_4 / acid / H^+ **AND** reflux; [1 mark]

Identification of correct reagent and one reaction condition scores [1 mark]

Identification of correct reagent and both reaction conditions scores [2 marks]

iii) The type of reaction that occurs in reaction **2** is:

- Hydrolysis; [1 mark]

iv) The role of NaBH_4 in reaction **4** is:

- A reducing agent; [1 mark]

v) A chiral centre means:

- It has a carbon / C / atom attached to four different groups / atoms / chains
OR
It has no plane / line of symmetry / has non-super(im)posable images; [1 mark]

[Total: 8 marks]

- The reaction in part **(i)** is the reaction between an acid and a metal carbonate which will produce a salt (product **Y**), carbon dioxide and water
- Reaction **1** shows the reaction of ethanal to form 2-hydroxypropanenitrile
 - Aldehydes or ketones undergo nucleophilic addition when reacted with HCN
 - HCN is poisonous so is formed in situ (in the reaction vessel) when KCN and dilute sulfuric acid is heated
 - This is a very useful reaction as it gives a means of increasing the length of the

carbon chain

- In reaction **2**, the hydroxynitrile forms a carboxylic acid
 - Hydrolysis of nitriles with either dilute acid or dilute alkali followed by acidification are methods of preparation of a carboxylic acid that you should know
- In reaction **3**, oxidation of a primary alcohol to a carboxylic acid and a secondary alcohol to a ketone
- In reaction **4**, focus on what part of the reactant is changing: the -COOH remains the same but the C=O double bond becomes C-OH , i.e an alcohol has formed
 - NaBH_4 is used as a reducing agent to reduce aldehydes and ketones to produce alcohols
 - Aldehydes are reduced to primary alcohols
 - Ketones are reduced to secondary alcohols
- This question shows you how familiar you need to be with all of the different types of reactions, reagents and conditions for a wide range of organic compounds
- The first step is to identify the changes that occur to the functional groups from the reactants to the products, then recall the reaction conditions necessary for those changes to occur

2a

i) The isomers are:

- A = 3,3-hydroxymethylbutanoic acid **OR** 3-hydroxy-3-methylbutanoic acid;
AND
B = 2,3-hydroxymethylbutanoic acid **OR** 2-hydroxy-3-methylbutanoic acid; [1 mark]

ii) A reagent to distinguish between the isomers and the type of reaction is:

- Acidified / H^+ / H_2SO_4
AND
(potassium) dichromate / $\text{K}_2\text{Cr}_2\text{O}_7$ / $\text{Cr}_2\text{O}_7^{2-}$
OR
Acidified / H^+ / H_2SO_4
AND
(potassium) manganate(VII) / KMnO_4 / MnO_4^- ; [1 mark]
- The reaction is oxidation; [1 mark]

iii) The observations in each case are:

Example 1 (with potassium dichromate)

- Isomer A has no change / no reaction; [1 mark]
- Isomer B changes the colour from orange to green; [1 mark]

OR

Example 2 (with potassium manganate(VII))

- Isomer A has no change / no reaction; [1 mark]
- Isomer B changes the colour from purple to colourless; [1 mark]

[Total: 5 marks]

- Where there are two functional groups present naming priority is generally given to the functional group with the larger mass
 - In this case the carboxylic acid takes priority over the hydroxyl group
 - Carbon groups are put together in the carbon 'skeleton' so the methyl group comes in front of the butanoic acid, not the hydroxyl group
- You need to know the test for an alcohol and the result, although the test will not work on all classes of alcohol as this question shows
- You must also state when there is no reaction, not just the positive result of a test
- Before and after colours are essential.
 - E.g., stating that dichromate turns green would not get you the mark

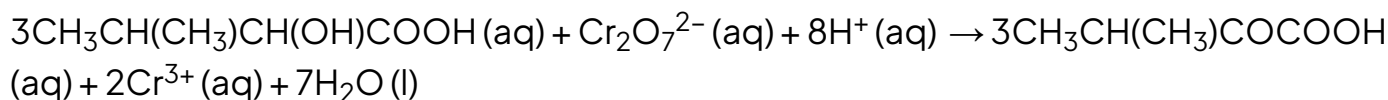
2b

b)

ii) The overall redox reaction is:

- $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}(\text{OH})\text{COOH} \rightarrow \text{CH}_3\text{CH}(\text{CH}_3)\text{COCOOH} + 2\text{H}^+ + 2\text{e}^-$; [1 mark]

ii) The overall redox reaction is:



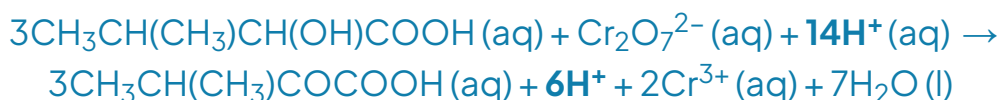
- The correct reactants and products are shown; [1 mark]
- Balancing the equation; [1 mark]

[Total: 3 marks]

- Compound B is a secondary alcohol so it can only be oxidized to a ketone
- This involves the loss of two hydrogens
 - This is not hydrogen gas, but hydrogen ions, which must be balanced by the addition of two electrons on the right-hand side of the half equation
- Always check that half-equations have balanced charges on either side
- To balance the half-equations, deduce the lowest common factor that balances the number of electrons
- The B to C half-equation has two electrons, so it must be multiplied by three, so that both half equations now have six electrons



- When you combine them together there are H^+ ions on each side which you can cancel down



- State symbols can be ignored in the equation

2c

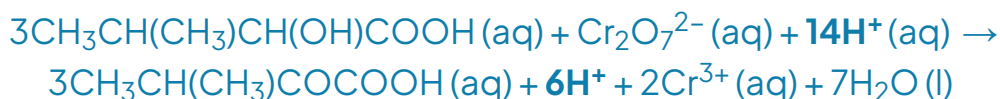
c) The reaction conditions can be changed in the oxidation of ethanol by:

- To produce ethanal: the reaction is carried out as a simple distillation, so the product is removed as soon as it is formed; [1 mark]

- Always check that half equations have balanced charges on either side
- To balance the half-equations, deduce the lowest common factor that balances the number of electrons
- The B to C half-equation has two electrons, so it must be multiplied by three, so that both half equations now have six electrons



- When you combine them together there are H^+ ions on each side which you can cancel down



- State symbols can be ignored in the equation

2c

c) The reaction conditions can be changed in the oxidation of ethanol by:

- To produce ethanal: the reaction is carried out as a simple distillation, so the product is removed as soon as it is formed; [1 mark]
- To produce ethanoic acid: the reaction is heated under reflux;

OR

An excess of the oxidising agent is used; [1 mark]

[Total: 2 marks]

- The reason you want to distil off aldehydes as soon as they are formed is so that you don't allow them to further oxidise to a carboxylic acid
- Aldehydes do not have the hydrogen bonds that are present in alcohols and carboxylic acids, so they have lower boiling points and vaporise more easily
- The reactions are equilibrium reactions, so removing the aldehyde product encourages the equilibrium to shift to the right in favour of producing more aldehyde, and so it increases the yield of the reaction

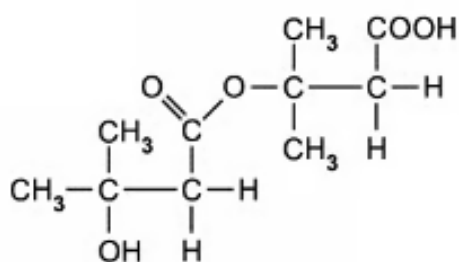
2d

d)

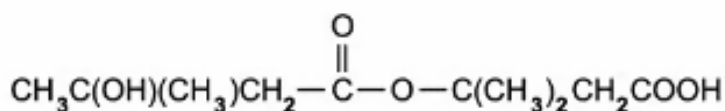
i) The type of reaction taking place and the reaction conditions are:

- Esterification / condensation; [1 mark]
- Warm / heat with concentrated sulfuric acid / H_2SO_4 ; [1 mark]

ii) The structure of the product is:



OR



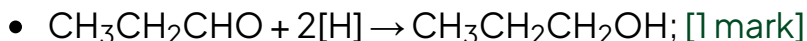
- Either correct drawing showing the ester group; [1 mark]

[Total: 3 mark]

- A dimer is any molecule made from two repeating units
 - Dimers are quite commonly found in sugars e.g., disaccharides like sucrose which is made from two glucose molecules joined together
- The role of the sulfuric acid is a catalyst for the reaction
- Water is also a product of the reaction, but don't get confused
 - The reaction is not a condensation reaction even though water is produced
 - Condensation refers to the joining together of two molecules to make one

3a

a) The equation for the reaction of propanal with NaBH_4 is:



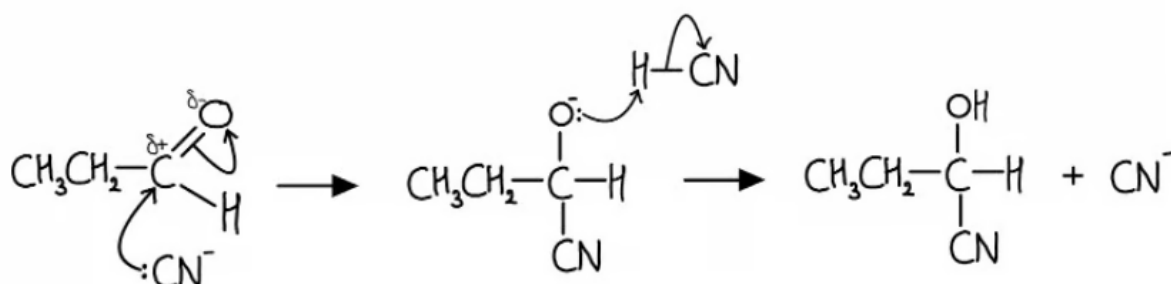
[Total: 1 mark]

- Aldehydes undergo reduction with NaBH_4 to form primary alcohols
- Ketones undergo reduction with NaBH_4 to form secondary alcohols
- You must use $[\text{H}]$ in the equation, not H_2
- These equations should also be balanced
- So if using $[\text{O}]$ for oxidation or $[\text{H}]$ for reduction you need to make sure the number of oxygen or hydrogen atoms each side of the equation is the same

3b

b) The specific type of isomerism shown by the product **Y** is:

- Optical isomerism; [1 mark]



- Curly arrow from carbon lone pair of CN^- to the C in the $\text{C}=\text{O}$ bond; [1 mark]
- Curly arrow from the $\text{C}=\text{O}$ bond to the O of the $\text{C}=\text{O}$ bond

AND

Dipole on the $\text{C}=\text{O}$ bond; [1 mark]

- Curly arrow from the lone pair of the O^- to an H; [1 mark]

- The $\text{C}=\text{O}$ is planar / flat

AND

$:\text{CN}^-$ can attack from above or below / either side (causing the optical isomers); [1 mark]

[Total: 5 marks]

- It can sometimes help to write the equation as a starting point, think of it as planning your mechanism answer out
 - For this mechanism, it makes it less likely that you will have the hydroxyl (-OH) and cyano (-CN) groups on the wrong carbon
- The hydrogen ion could also come from the water or the H_3O^+ ions present in the slightly acidic solution
 - You don't need to remember all of these
 - One equation is perfectly adequate.
- Although you may be allowed stereoisomerism in an exam in some cases, being more specific stating optical isomerism is better and ensures that you get the mark
 - In this case, the question asked for the specific type of isomerism shown, meaning that stereoisomerism would not be allowed for the mark

3c

c) The percentage yield for 5.00 cm^3 of propanal reacting to form 4.25 g of propanoic acid is:

- Moles of propanal = $\frac{5 \times 0.81}{58} = 0.0698 \text{ moles}$; [1 mark]
- Theoretical mass of propanoic acid = $0.0698 \times 74 = 5.167 \text{ g}$; [1 mark]
- Percentage yield calculation = $\frac{4.25}{5.167} \times 100 = 82.3\%$; [1 mark]

[Total: 3 marks]

- Remember to convert the volume of propanal into a mass in g before calculating the number of moles
- Answer should be given to 3 significant figures, as every piece of information given in the question is also given to 3 significant figures

3d

d)

i) The observations are:

- Orange precipitate; [1 mark]

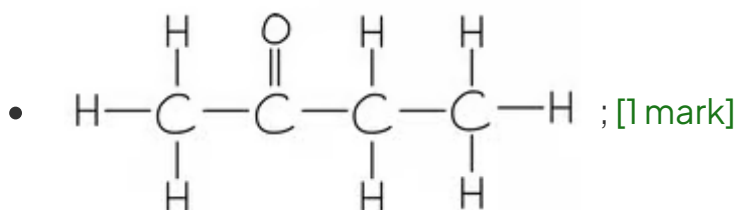
ii) The functional group present is:

- Ketone; [1 mark]

iii) Q is:

- Butanone

OR



[Total: 3 marks]

- 2,4-dinitrophenylhydrazine reacts with both aldehydes and ketones to give an orange precipitate
- Tollens' reagent will give a silver mirror with aldehydes, but not with ketones
 - Therefore compound Q is a ketone
- The M_r is 72 so determine the structure must be
 - $4 \times \text{C} = 48$
 - $8 \times \text{H} = 8$
 - $1 \times \text{O} = 16$
 - Total = 72

3e

e)

i) A suitable reducing agent is

- NaBH_4

OR

NaAlH_4

OR

LiAlH_4

OR

H_2 / Ni or Pt ; [1 mark]

ii) The functional group in **R** is:

- Secondary alcohol; [1 mark]

iii) The structural formula is:

- $\text{CH}_3\text{CH}_2\text{CHOHCH}_3$; [1 mark]

[Total: 3 marks]

- **Careful:** in part (ii) you have been asked for two words, so as well as stating that the functional group is an alcohol you must also state that it is a secondary alcohol
- Pay attention to the wording in the question
 - Structural formula is not the molecular formula ($\text{C}_4\text{H}_8\text{O}$)
 - If you gave this you would not gain the mark

4a

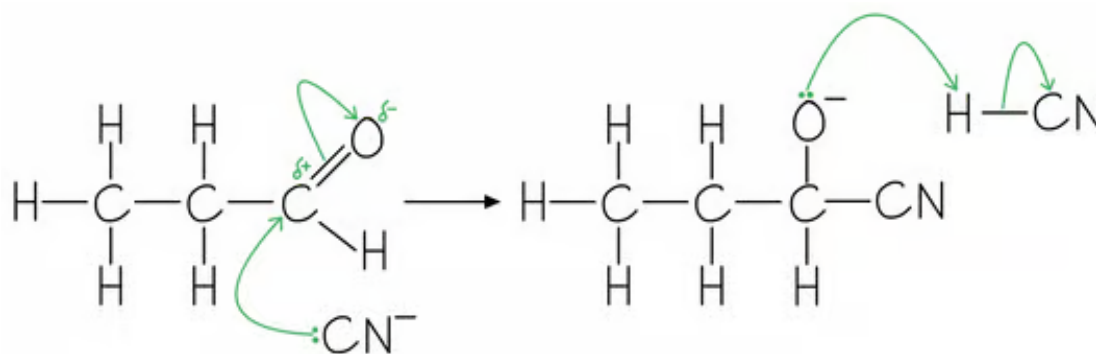
a)

i) The conditions required for reaction **2** are:

- Heat; [1 mark]
- KCN; [1 mark]
- (Dilute) sulfuric acid / catalyst; [1 mark]

ii) The completed mechanism is:

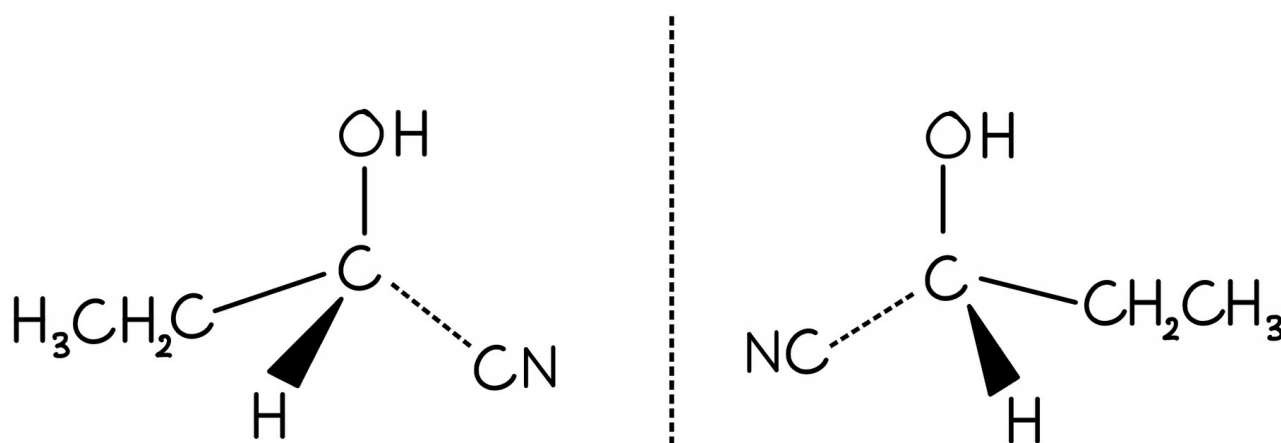
- Curly arrow from lone pair on C atom to δ^+ carbon atom; [1 mark]
- Curly arrow from C=O bond to δ^- oxygen atom; [1 mark]
- Curly arrow from lone pair on negatively charged oxygen atom to H atom in H-CN; [1 mark]
- Correct lone pairs and partial charges identified; [1 mark]

**[Total: 7 marks]**

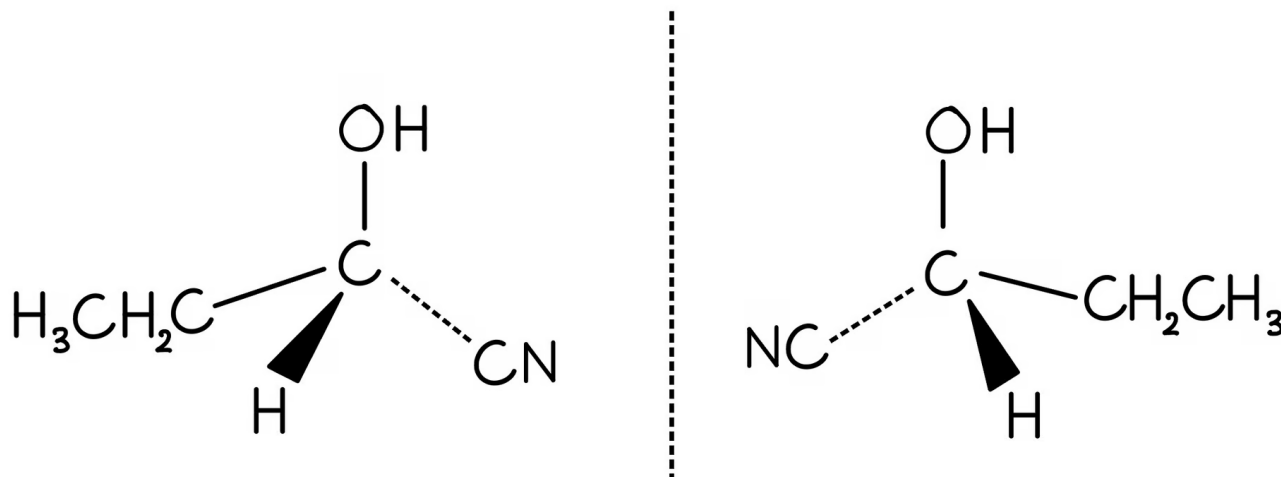
- The reagents and conditions for the formation of a hydroxynitrile from aldehyde or a ketone do not include HCN
 - This is because HCN, hydrogen cyanide, is poisonous so it is formed 'in situ' (in the reaction vessel) by the reaction of potassium cyanide and dilute sulfuric acid (or a catalyst)
- When completing mechanisms you are expected to include more than just curly arrows
 - Partial charges and lone pairs must also be included
- **Remember:** curly arrows must go from a negative area of charge to a positive area of charge

4b

b) The two optical isomers are:



b) The two optical isomers are:



[Total: 2 marks]

- A chiral center is a carbon atom that has four different atoms or groups of atoms attached
- In 2-hydroxybutanenitrile, carbon 2 is attached to H, OH, CN and CH₂CH₃
- When drawing the optical isomers, make sure you draw wedges to show them in three dimensions

4c

c) The reagent(s) for reaction **3** are:

- Dilute acid;
OR
Dilute alkali
AND
(followed by acidification by) sulfuric acid / hydrochloric acid; [1 mark]

[Total: 1 mark]

- Nitriles are hydrolysed by either dilute acid or dilute alkali
 - Hydrolysis by dilute acid results in the formation of a carboxylic acid and ammonium salt
 - Hydrolysis by dilute alkali results in the formation of a sodium carboxylate salt and ammonia
 - Acidification by a strong acid such as sulfuric acid or hydrochloric acid is

required to change the carboxylate ion into a carboxylic acid

4d

d) The structural formula for compound **A** is:

- $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$; [1 mark]

[Total: 1 mark]

- Acidified potassium dichromate will oxidise primary alcohols to aldehydes
- Therefore compound **A** must be a primary alcohol
- The colour change observed in this reaction is orange to green- you could be asked this so make sure you know it!

4e

e) The positive results are:

Reagent	Observation for positive result	Compounds that will give a positive result
I_2 / NaOH (aq)	Yellow precipitate formed; [1 mark]	Propanol AND Propanal; [1 mark]
Na_2CO_3	Fizzing / bubbles of gas; [1 mark]	None; [1 mark]
2,4-DNPH	Orange precipitate formed; [1 mark]	Propanal; [1 mark]

[Total: 6 marks]

- Both alcohols containing an $\text{R-CH}_2\text{OH}$ and aldehydes group will give a positive result when reacted with tri-iodomethane (iodoform test)
- 2,4-DNPH (2,4-dinitrophenylhydrazine) tests for the presence of a C=O bond
 - Aldehydes and ketones undergo a condensation reaction with 2,4-DNPH (2,4-dinitrophenylhydrazine)- the compound formed is an orange precipitate
 - Alcohols do not give a positive result for this reaction
- The reaction with Na_2CO_3 would give a positive result with a carboxylic acid group